Sahaj Bhadja

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Summary

Engineering Applications Software Engineer with 2+ years building **C++/Python systems**, **Unix data pipelines**, automated unit tests, and **traceable validation workflows** using networking, debugging, and technical documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **system validation**, reconciliation, error handling, cloud storage integration, CI/CD releases, and stable production monitoring workflows reliably at enterprise scale.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured test workflows**, model switching, fallback paths, persistent results, and reviewer controls supporting repeatable analytical processing for technical users.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building an **algorithmic Python pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through **system-level debugging** of HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interactions across diverse autonomous-driving simulation scenarios reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Increased pipeline efficiency by 68% through **JSON, CSV, and Parquet processing**, normalizing heterogeneous inputs with AWS Glue and S3 while reducing storage costs by 15% across repeatable data workflows.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship | *India*
- Reduced centralized-control and quantum-risk exposure by 25% through a **novel consensus algorithm**, comparing benchmark behavior, performance trade-offs, and failure characteristics against established distributed mechanisms systematically and reproducibly.
- Published peer-reviewed **systems research** and presented at IISc Bengaluru through IEEE venues, documenting architecture, simulation methodology, threat models, experimental trade-offs, and reproducible findings for technical and non-technical audiences.

Skills

Languages: C++, Python, Java, SQL, Bash, JavaScript
Data Formats & Networking: JSON, XML, CSV, Parquet, TCP/IP, Socket Programming, Client-Server Architecture
Systems & Architecture: Unix, Linux, Windows, macOS, Operating Systems, Multithreading, Concurrent Programming, Distributed Systems, CPU Scheduling, System Design, Object-Oriented Programming
Testing & Reliability: Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Debugging, Root-Cause Analysis, Fault Isolation, Performance Optimization, Monitoring, Data Validation
Data & Applications: PostgreSQL, Data Pipelines
Tools & Delivery: Git, GitHub, Jenkins, Docker, Kubernetes, CI/CD, Configuration Management, Application Packaging, Technical Documentation, Requirements Traceability, Agile

Projects

Telemetry Stream Processing and Anomaly Replay Harness - Design Concept
- Designed a **telemetry stream-processing harness** concept in C++ and Python with TCP/IP socket clients, XML/JSON parsers, timestamped frame logs, data-loss injection, anomaly replay, performance profiling, and deterministic regression tests.

Change Control and Test Evidence Workbench - Design Concept
- Designed a **traceable engineering workbench** concept linking change requests, requirements, anomaly reports, design inputs, configuration baselines, unit-test evidence, release packages, operations guidance, review history, and corrective actions.

CPU-Scheduling Simulator | Live
- Built a **C++ systems simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across operational scenarios.